

**Laboratorio de Microeconomía I**  
Centro de Investigación y Docencia Económicas  
Maestría en Economía - 2025  
Laboratorista: Arturo López  
Tarea 10

## Exercises

---

**Ex. 2.26** Let  $u(w) = -(b - w)^c$ . What restrictions on  $w, b$ , and  $c$  are required to ensure that  $u(w)$  is strictly increasing and strictly concave? Show that under those restrictions,  $u(w)$  displays *increasing* absolute risk aversion.

---

**Ex. 2.27** Show that for  $\beta > 0$ , the VNM utility function  $u(w) = \alpha + \beta \ln(w)$  displays decreasing absolute risk aversion.

---

**Ex. 2.28** Let  $u(x_1, x_2) = \ln(x_1) + 2 \ln(x_2)$ . If  $p_1 = p_2 = 1$ , will this person be risk loving, risk neutral, or risk averse when offered gambles over different amounts of income?

---

**Ex. 2.35** An investor must decide how much of initial wealth  $w$  to allocate to a risky asset with unknown rate of return  $r$ , where each outcome  $r_i$  occurs with probability  $p_i, i = 1, \dots, n$ . Using the framework of Example 2.6, prove that if the investor's preferences display *increasing* absolute risk aversion, the risky asset must be an 'inferior' good.

---

**Ex. 6.C.1<sup>B</sup>** Consider the insurance problem studied in Example 6.C.1. Show that if insurance is not actuarially fair (so that  $q > \pi$ ), then the individual will not insure completely.

---

**Ex. 6.C.15<sup>A</sup>** Assume that, in a world with uncertainty, there are two assets. The first is a riskless asset that pays 1 dollar. The second pays amounts  $a$  and  $b$  with probabilities  $\pi$  and  $1 - \pi$ , respectively. Denote the demand for the two assets by  $(x_1, x_2)$ .

Suppose that a decision maker's preferences satisfy the axioms of expected utility theory and that he is a risk averter. The decision maker's wealth is 1, and so are the prices of the assets. Therefore, the decision maker's budget constraint is given by

$$x_1 + x_2 = 1, \quad x_1, x_2 \in [0, 1].$$

- (a) Give a simple *necessary* condition (involving  $a$  and  $b$  only) for the demand for the riskless asset to be strictly positive.
- (b) Give a simple *necessary* condition (involving  $a, b$ , and  $\pi$  only) for the demand for the risky asset to be strictly positive.

---

In the next three parts, assume that the conditions obtained in (a) and (b) are satisfied.

- (c) Write down the first-order conditions for utility maximization in this asset demand problem.
  - (d) Assume that  $a < 1$ . Show by analyzing the first-order conditions that  $dx_1/da \leq 0$ .
  - (e) Which sign do you conjecture for  $dx_1/d\pi$ ? Give an economic interpretation.
  - (f) Can you prove your conjecture in (e) by analyzing the first-order conditions?
- 

**Ex (6.C.18<sup>B</sup>).** Suppose that an individual has a Bernoulli utility function  $u(x) = \sqrt{x}$ .

- (a) Calculate the Arrow–Pratt coefficients of absolute and relative risk aversion at the level of wealth  $w = 5$ .
  - (b) Calculate the certainty equivalent and the probability premium for a gamble  $(16, 4; \frac{1}{2}, \frac{1}{2})$ .
  - (c) Calculate the certainty equivalent and the probability premium for a gamble  $(36, 16; \frac{1}{2}, \frac{1}{2})$ . Compare this result with the one in (b) and interpret.
-